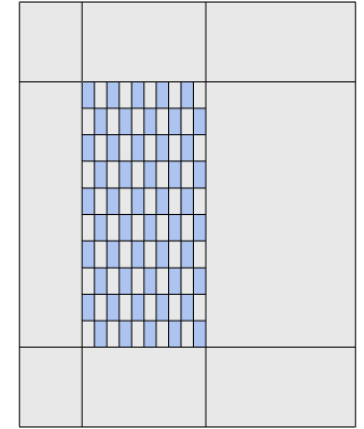
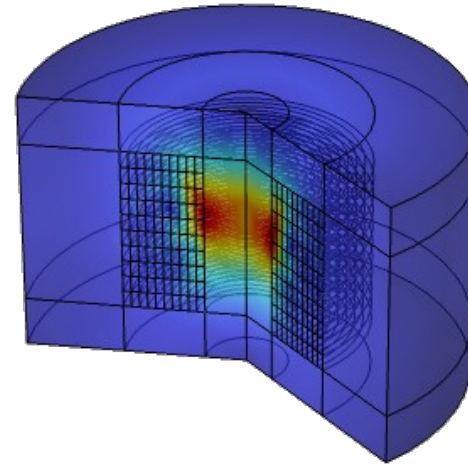


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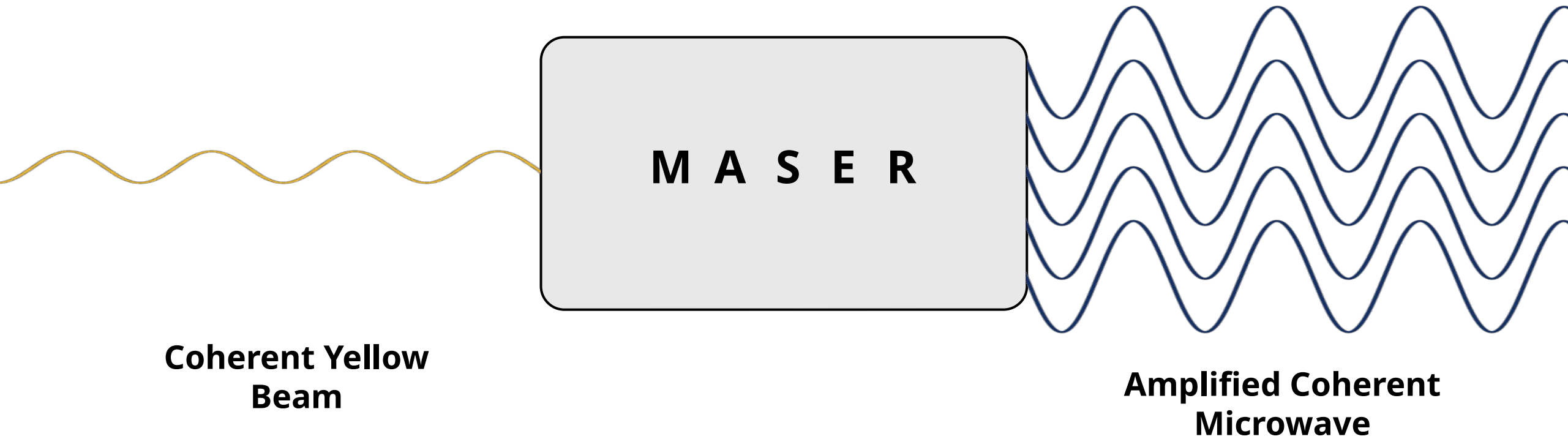
Quantum Design and Simulation of MASERs in the Strong Coupling Limit

Morgan Evans

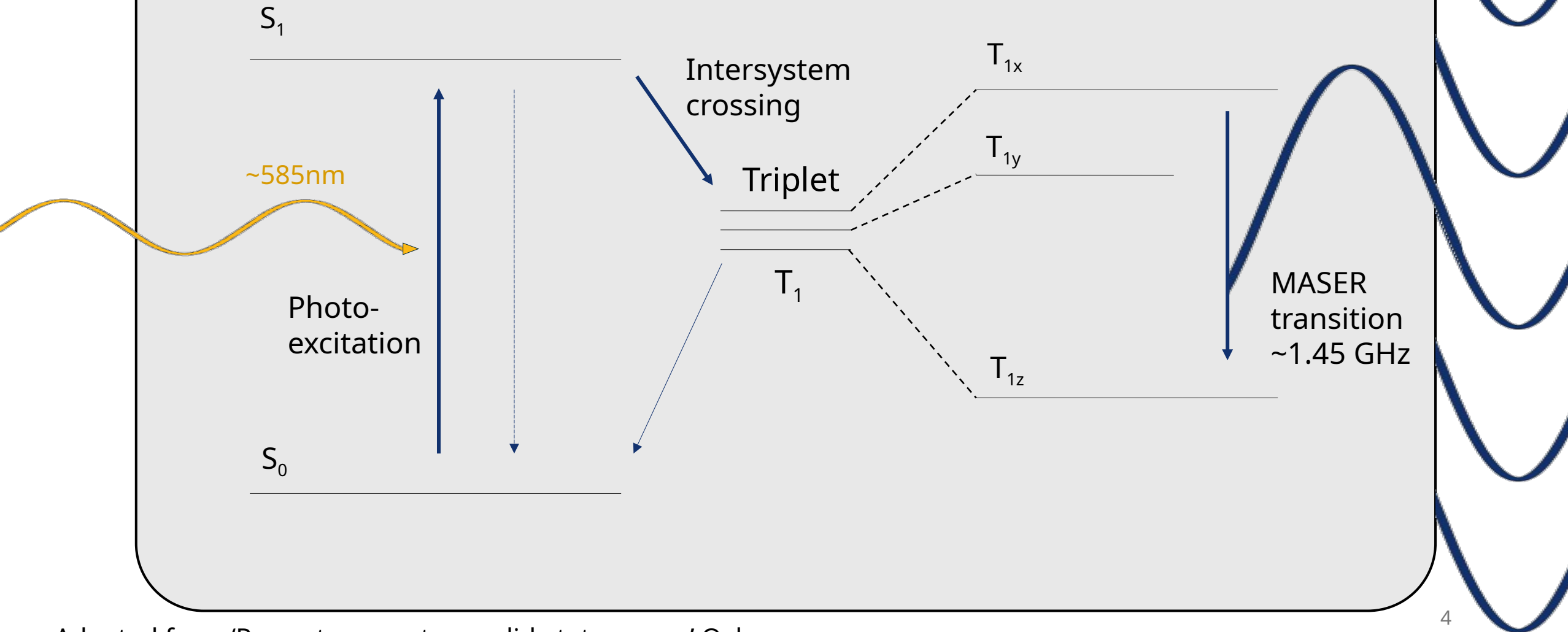
With help from Prof. Mark Oxborrow

Background

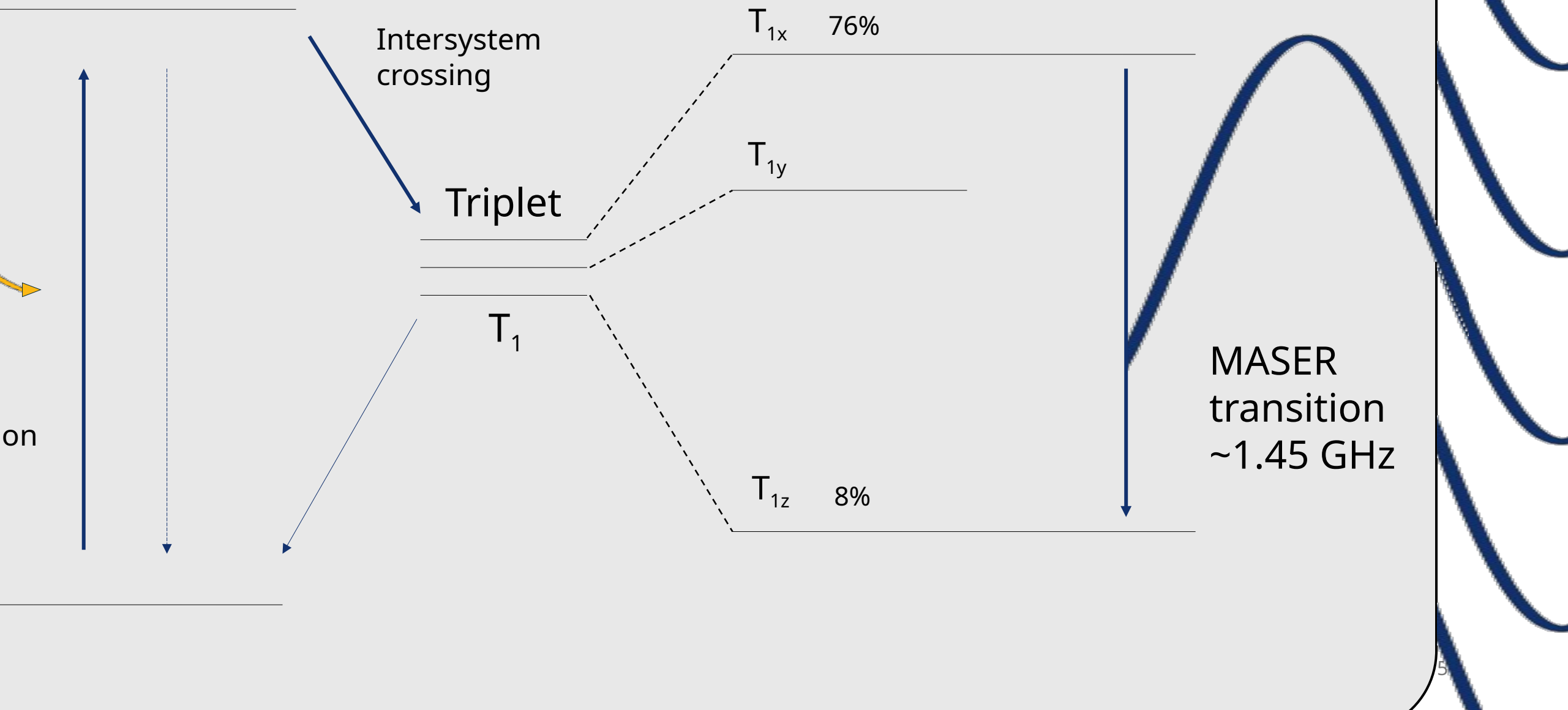
What is a MASER?



Microwave Amplification by Stimulated Emission of Radiation



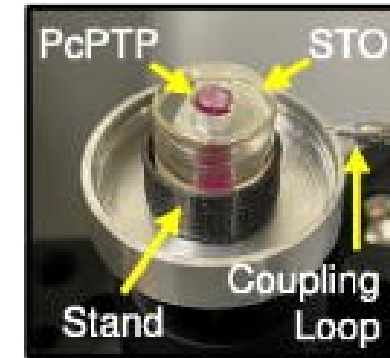
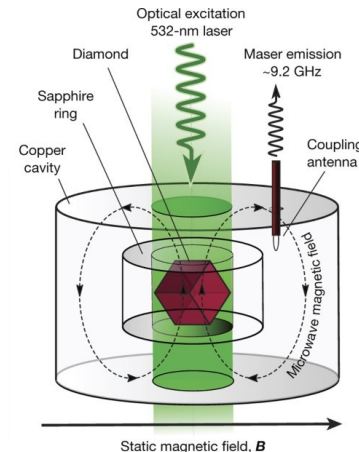
microwave Amplification by Stimulated Emission of Radiation



What is the problem?



First functioning MASER



1954^[1]

2012^[2]

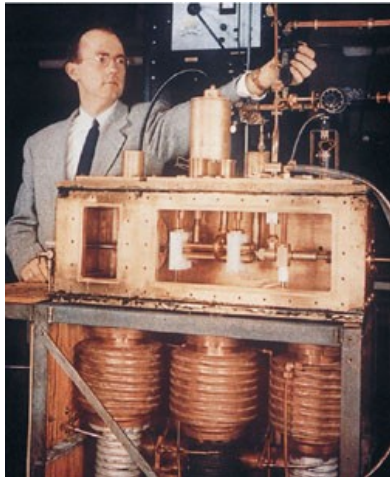
2018^[3]

2024^[4]

- [1] <https://physics.aps.org/story/v15/st4>
[2] <https://newatlas.com/solid-state-maser/23750/>
[3] <https://doi.org/10.1038/nature25970>
[4] <https://doi.org/10.1063/5.0181318>

What is the problem?

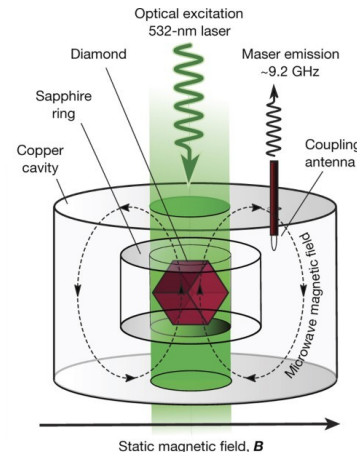
First room-temperature MASER.



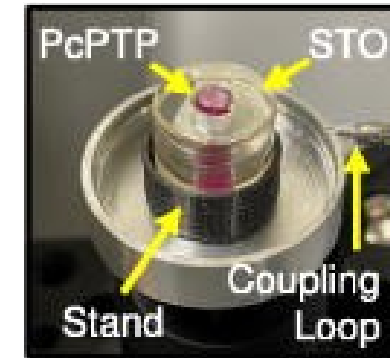
1954^[1]



2012^[2]



2018^[3]



2024^[4]

- [1] <https://physics.aps.org/story/v15/st4>
- [2] <https://newatlas.com/solid-state-maser/23750/>
- [3] <https://doi.org/10.1038/nature25970>
- [4] <https://doi.org/10.1063/5.0181318>

What is the problem?

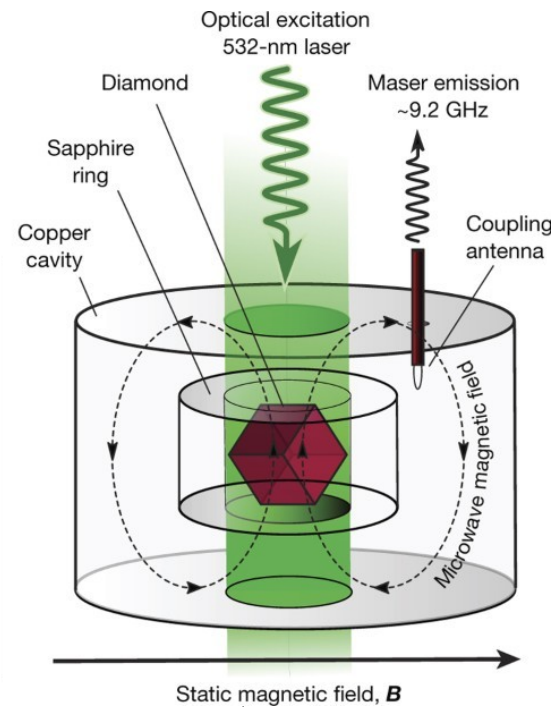
Continuous room-temperature MASER



1954^[1]

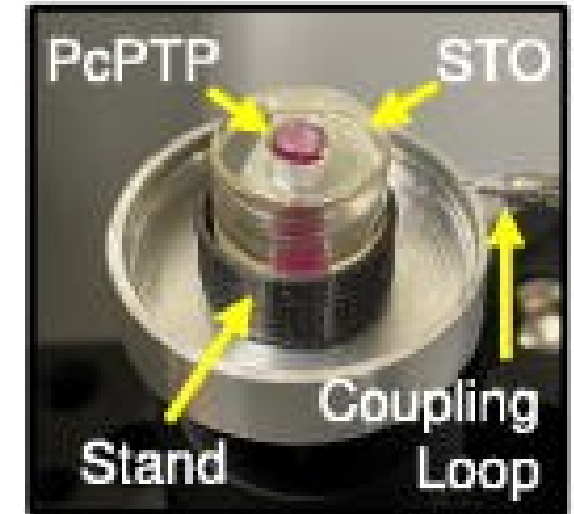


2012^[2]



2018^[3]

MASER-in-a-shoebox



2024^[4]

- [1] <https://physics.aps.org/story/v15/st4>
- [2] <https://newatlas.com/solid-state-maser/23750/>
- [3] <https://doi.org/10.1038/nature25970>
- [4] <https://doi.org/10.1063/5.0181318>

Different shape? A “beautiful” resonator?

Purcell Factor

$$= \frac{Q_f}{V_m}$$



High co-operativity



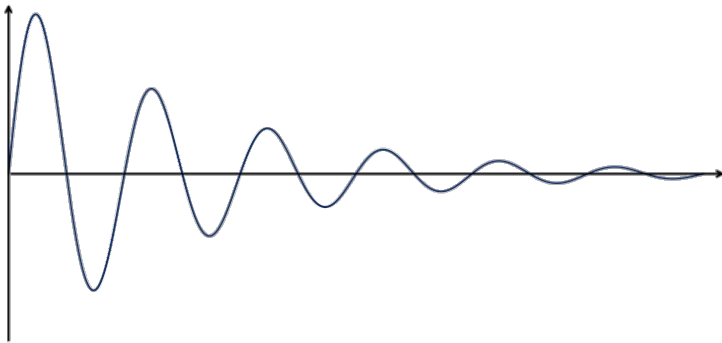
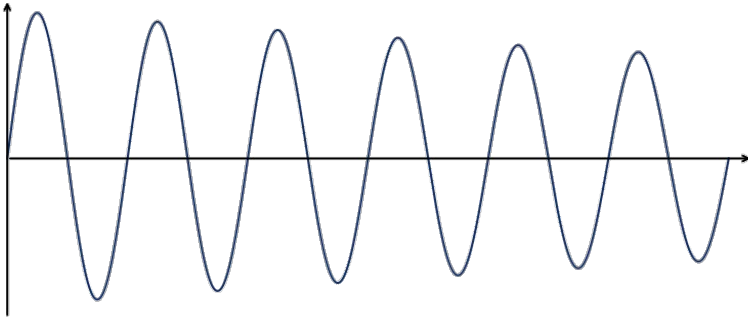
Low noise

Different shape? A “beautiful” resonator?

Purcell Factor

Q_f

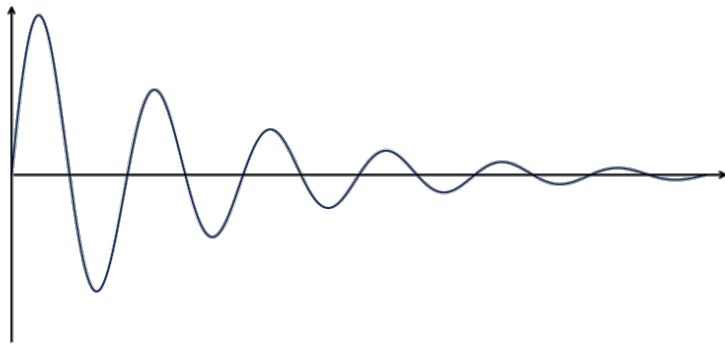
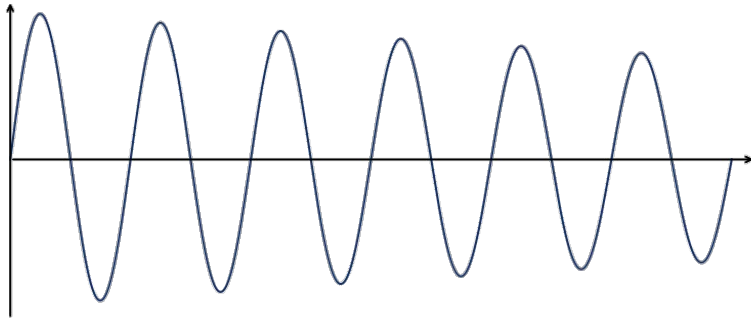
$$= \frac{1}{\nu_m}$$



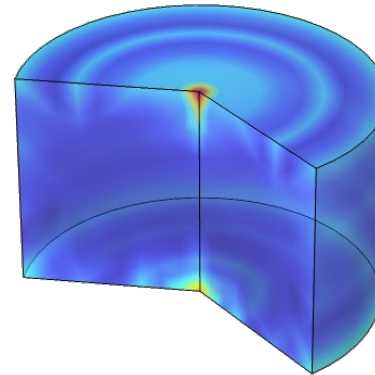
Different shape? A “beautiful” resonator?

Purcell Factor

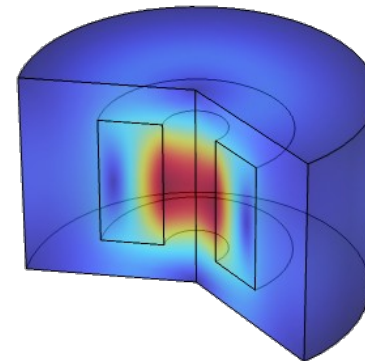
Q_f



V_m



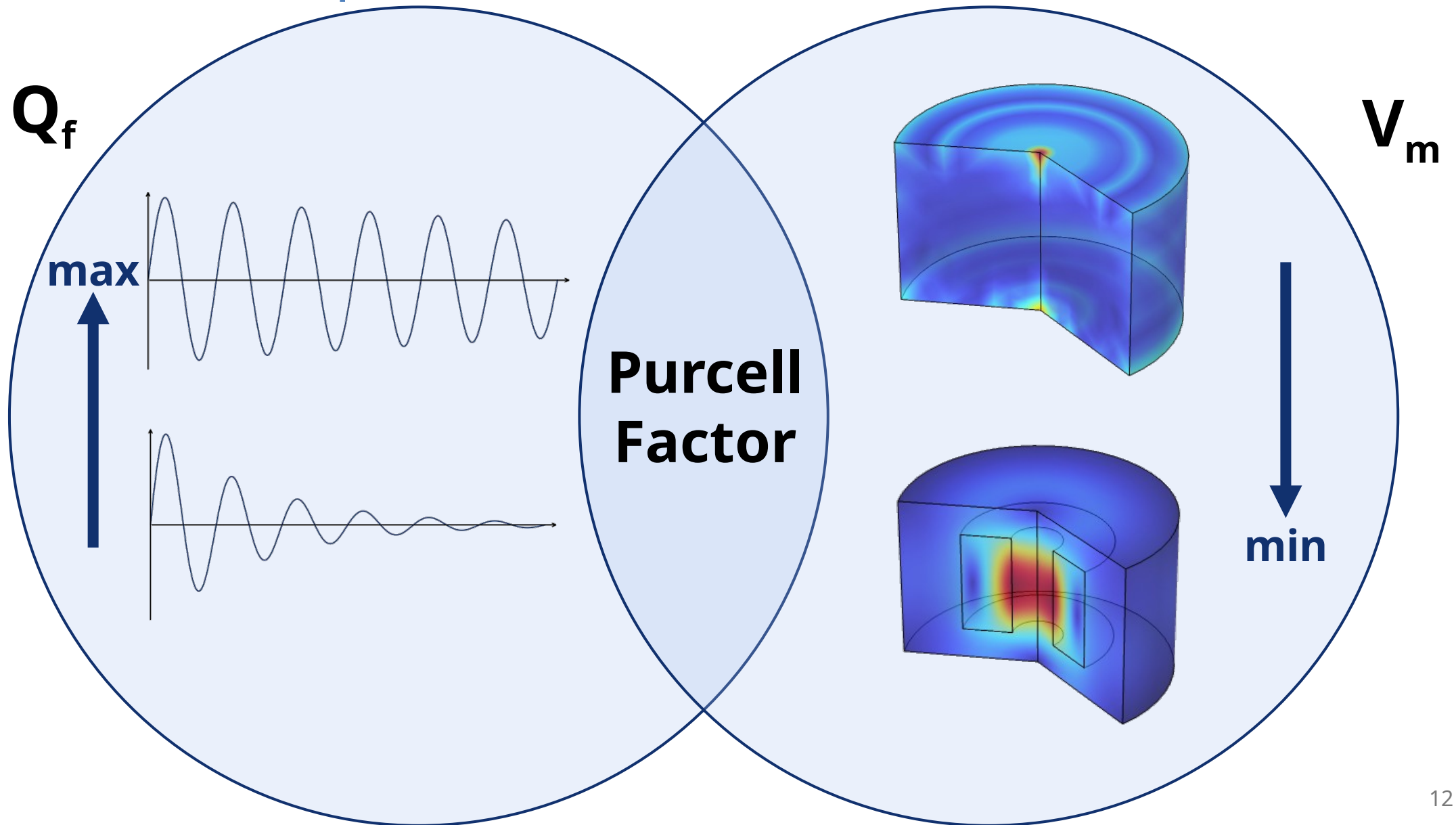
*Heatmaps of magnetic flux
plotted in COMSOL*



Flux
~~capacitors~~
'concentrators'

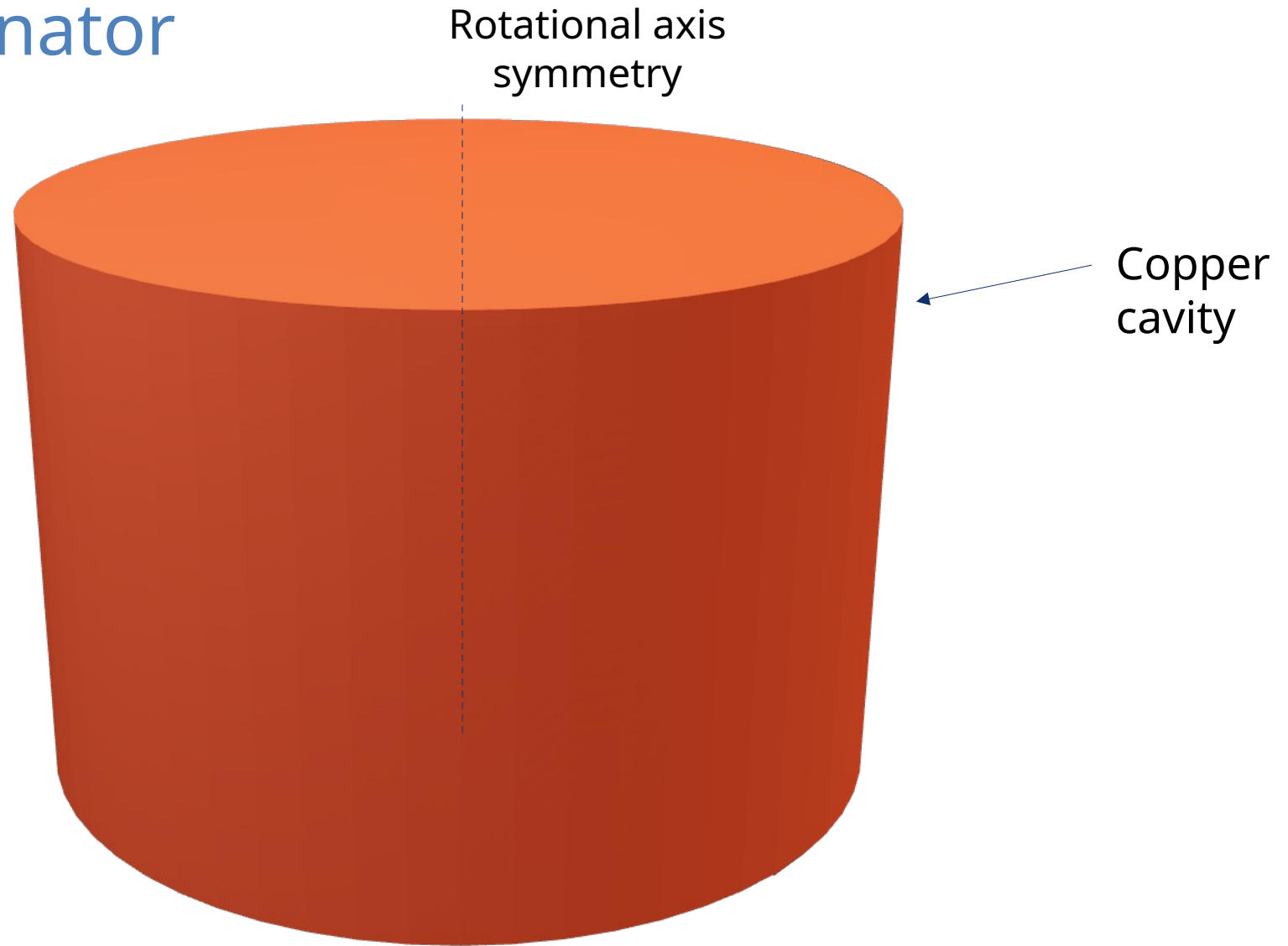


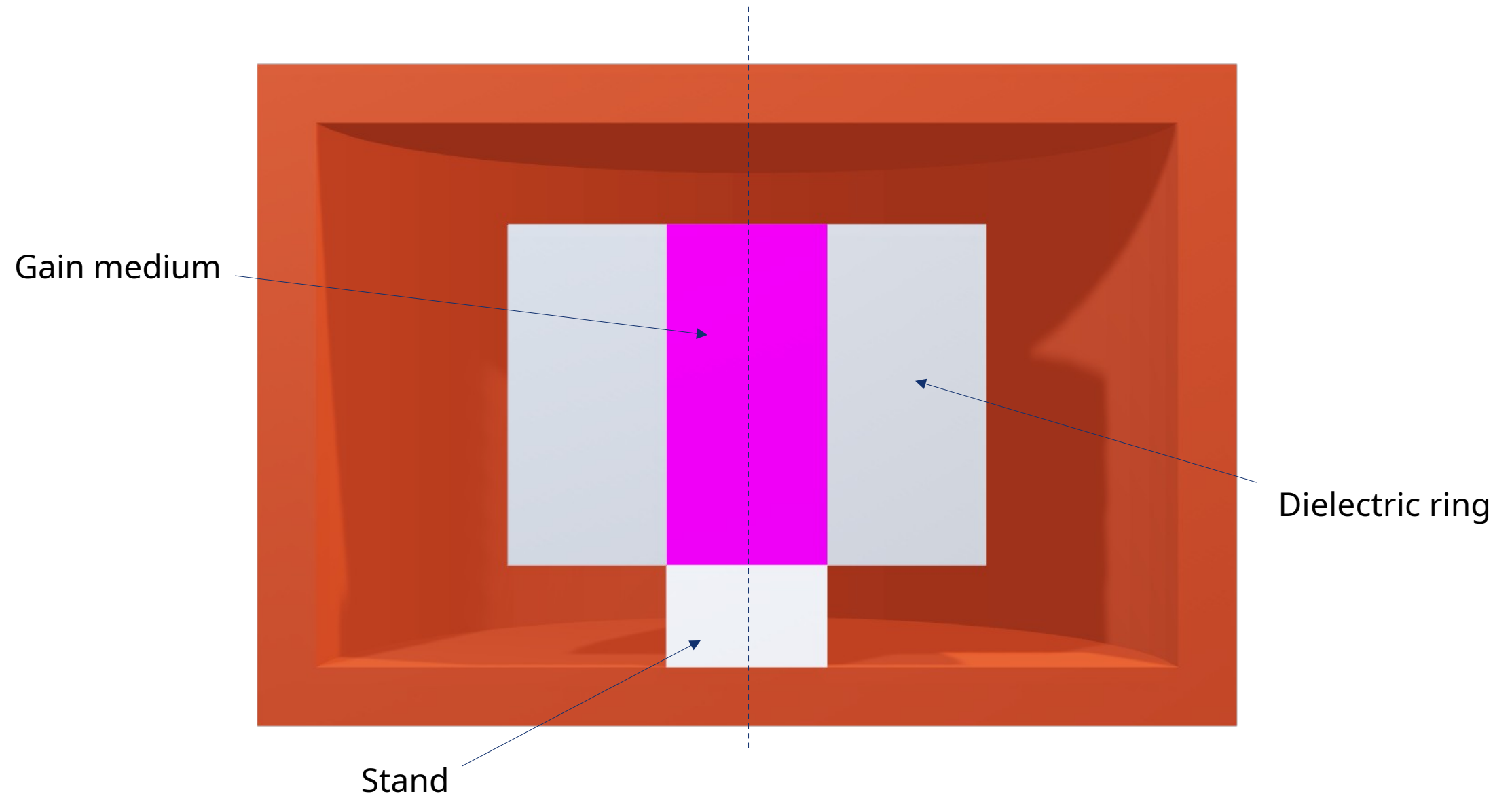
Different shape? A “beautiful” resonator?



What did I do?

A typical resonator

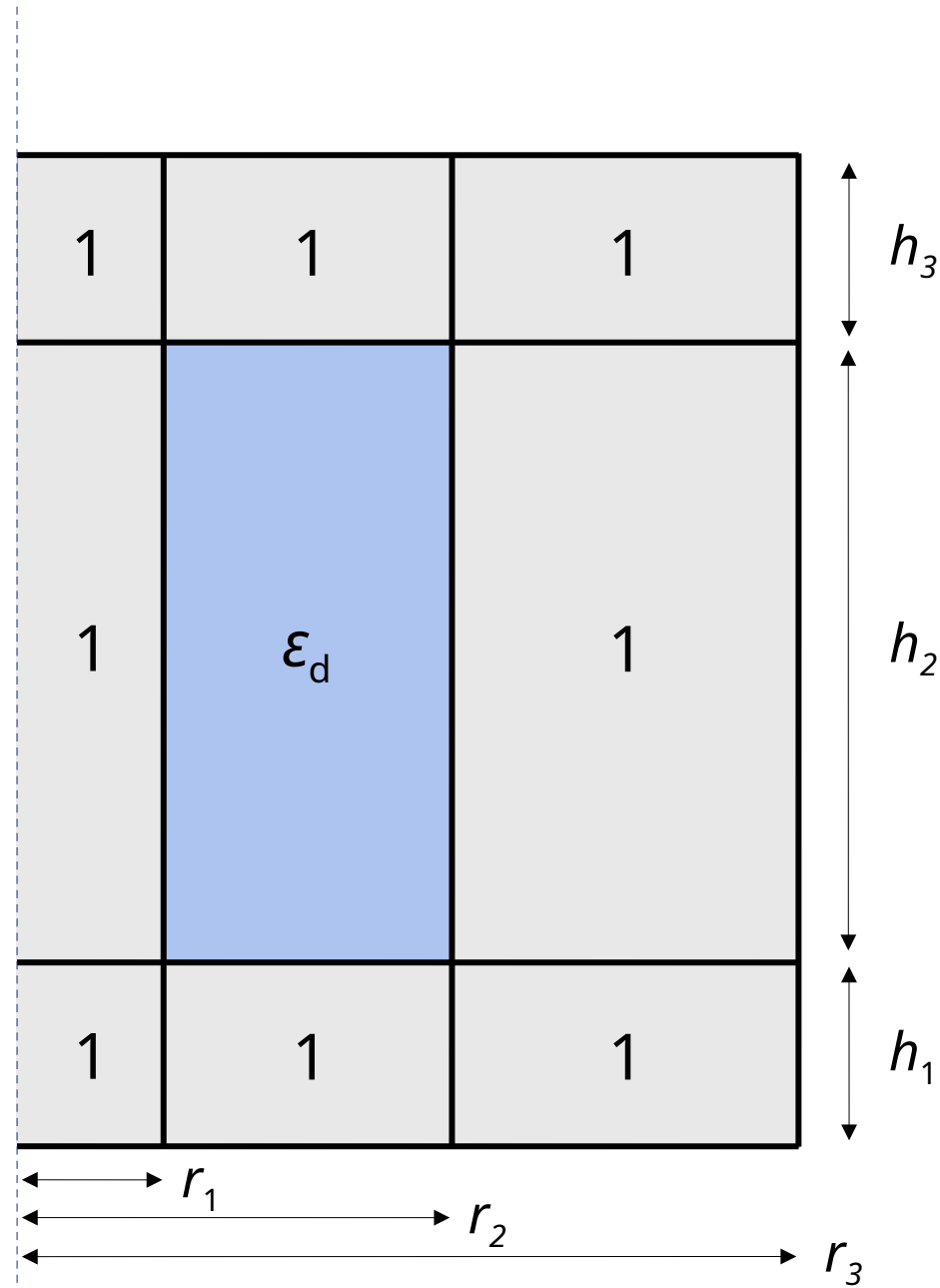


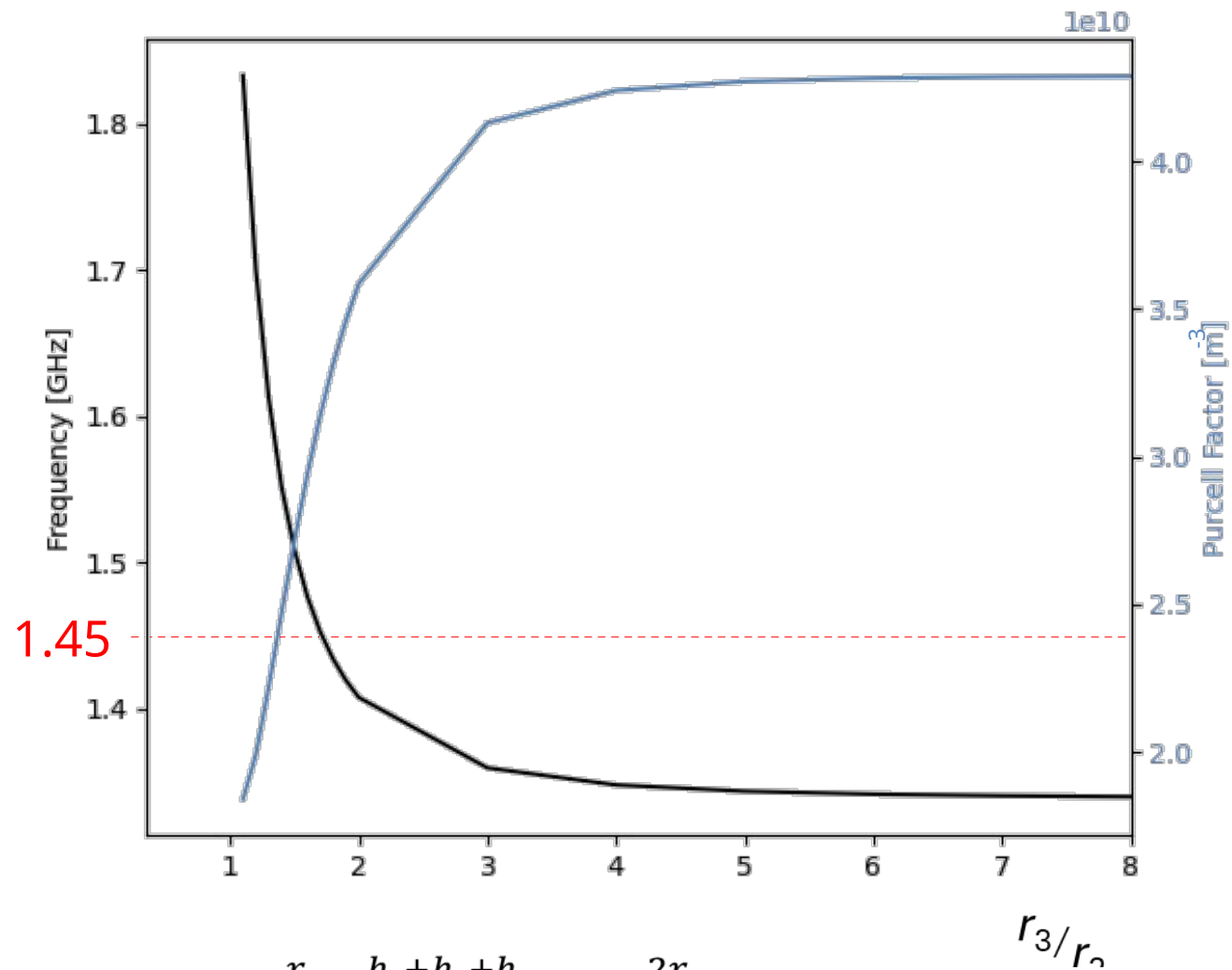
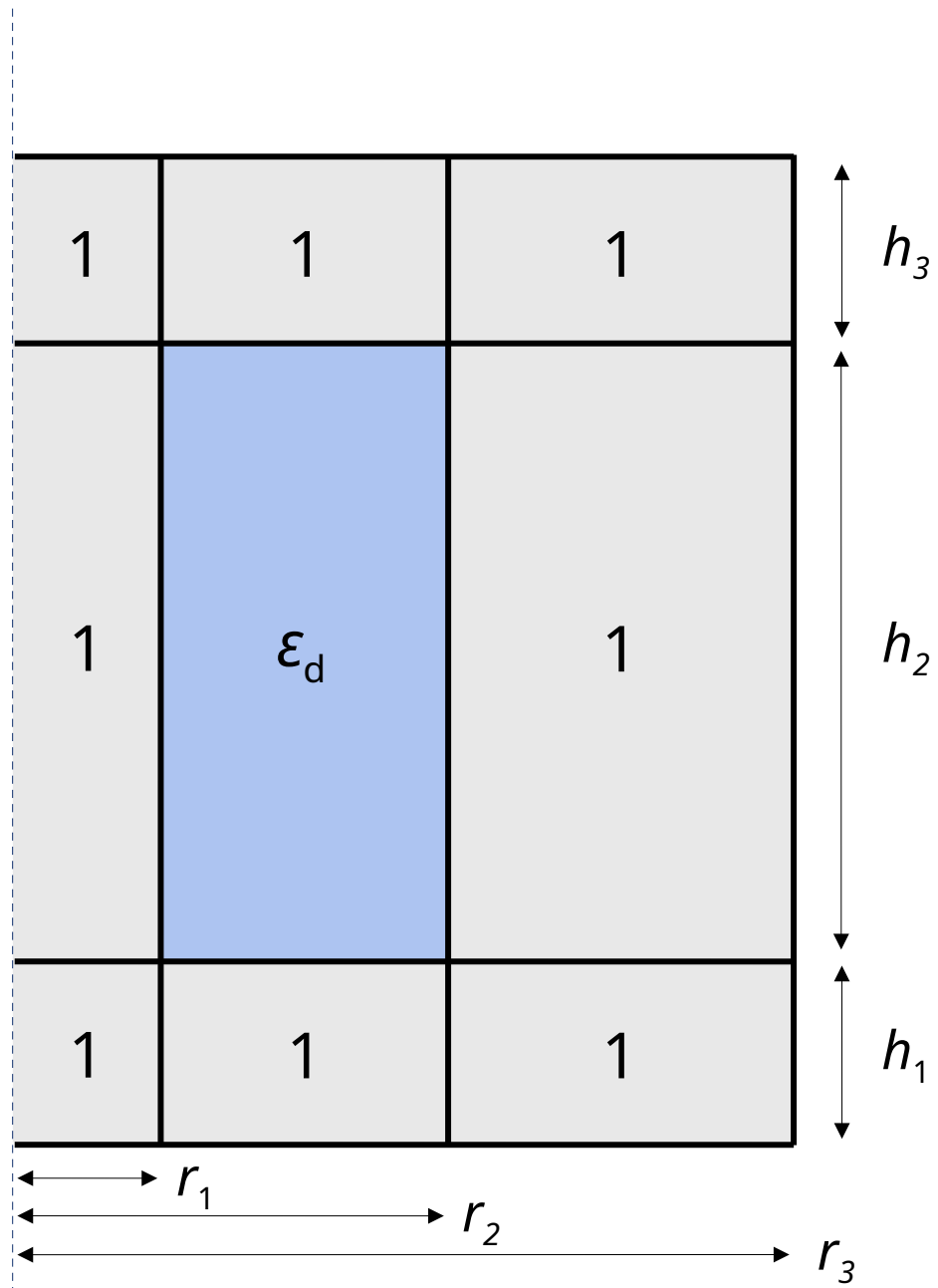


3x3 Space Grid

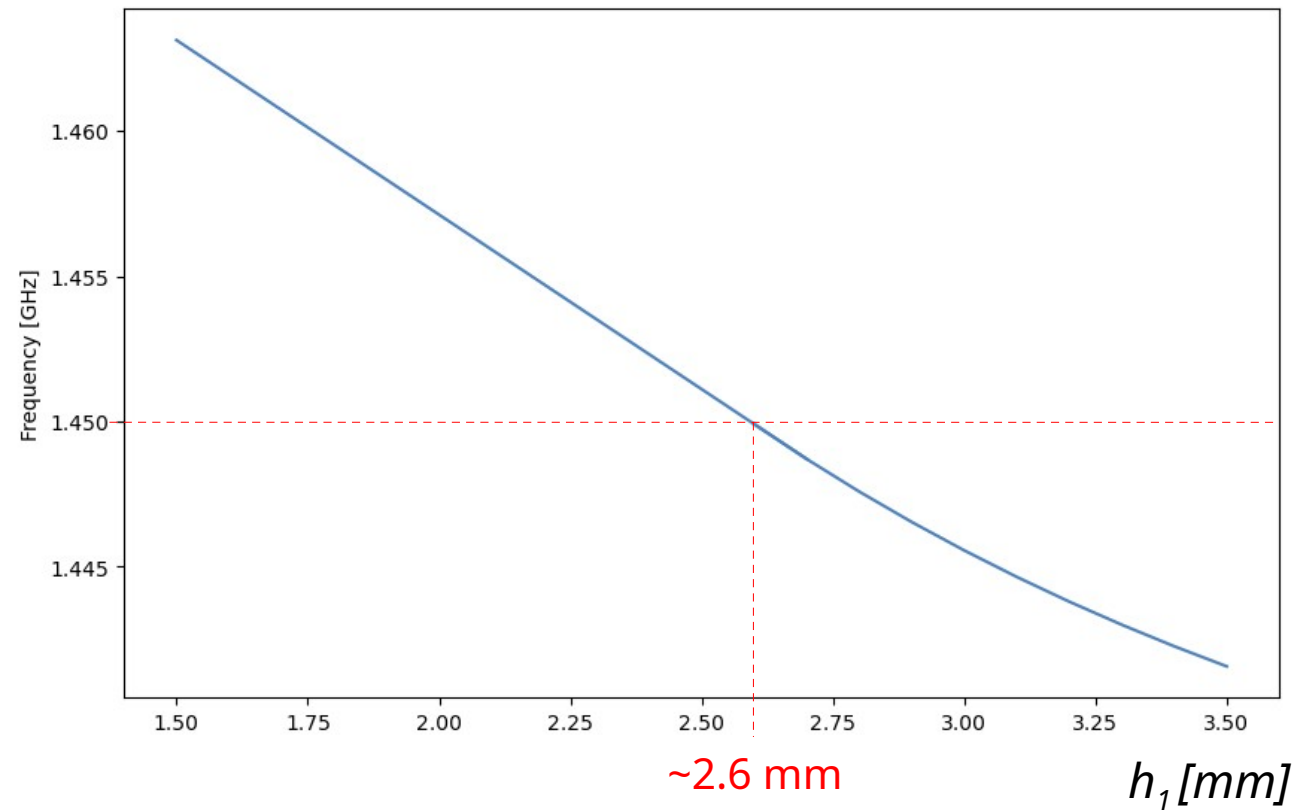
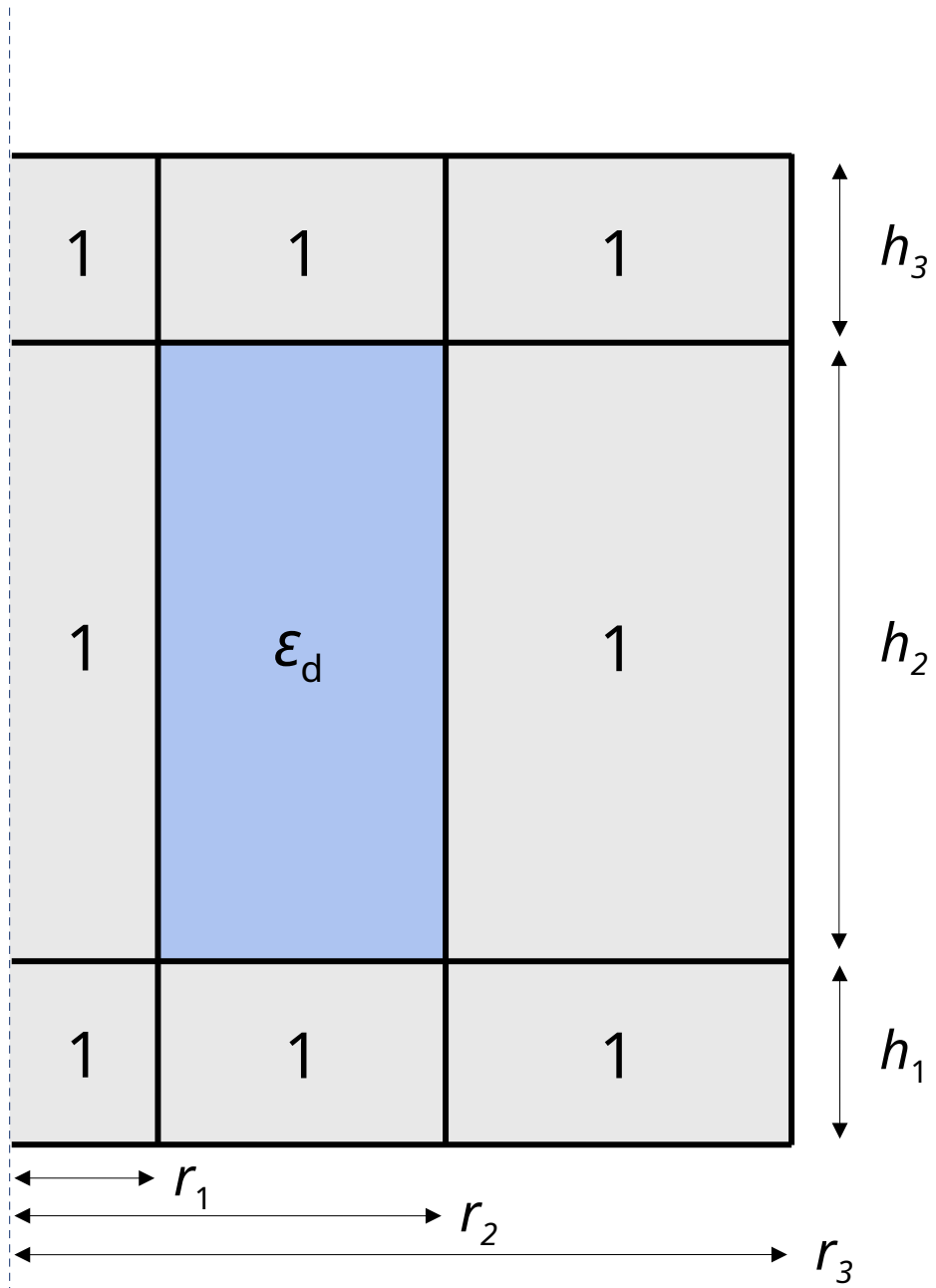
$$\epsilon_d = \begin{bmatrix} \epsilon_{||} & 0 & 0 \\ 0 & \epsilon_{||} & 0 \\ 0 & 0 & \epsilon_{\perp} \end{bmatrix}$$

Strontium titanate
 $\epsilon_d \approx 316$





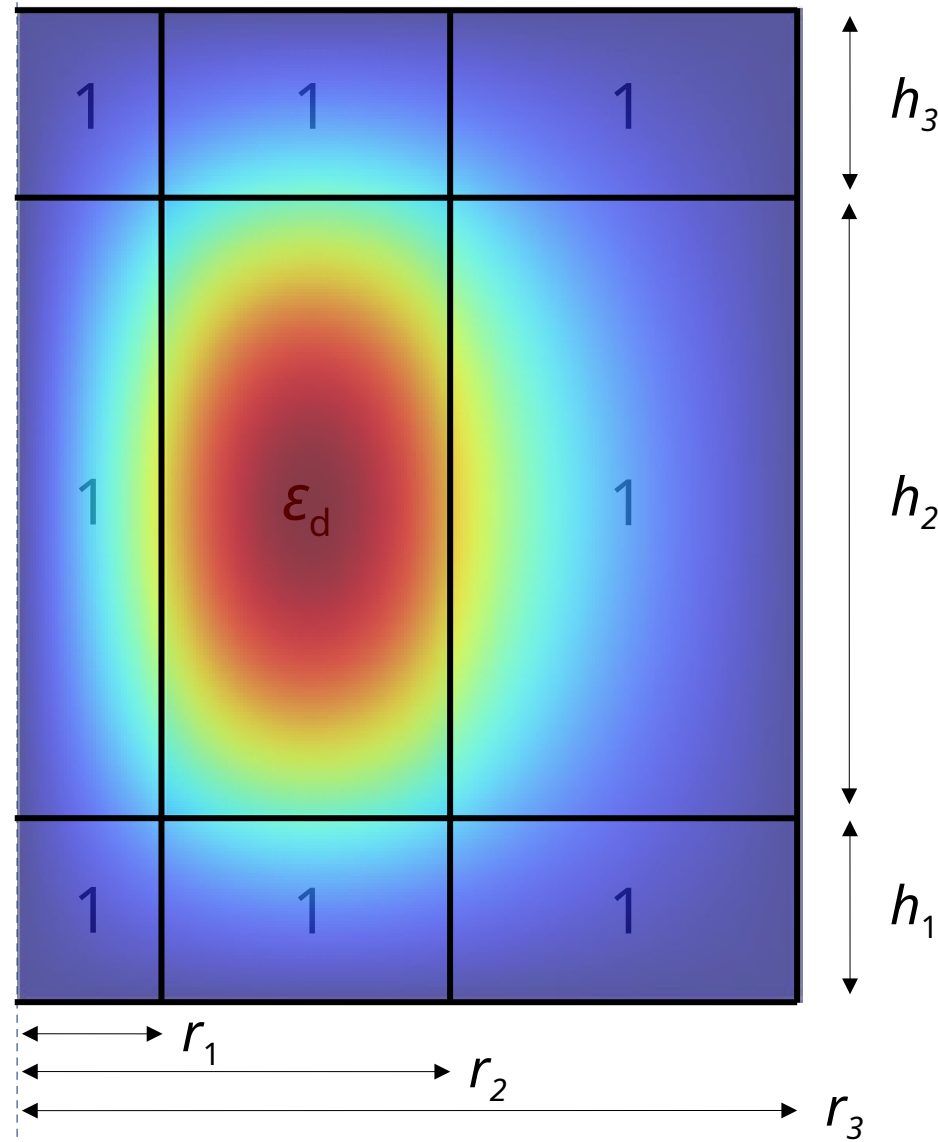
When $\frac{r_3}{r_2} = \frac{h_1 + h_2 + h_3}{h_2}$, and $\frac{2r_2}{h_2} = 1$



Where $h_1 = h_3$

Eigenfrequency=1.4499+1.1069E-4i GHz Surface: Electric field norm (V/m)

$$\begin{aligned}r_1 &= 2.05 \text{ mm}^{[1]} \\r_2 &= 6.1 \text{ mm} \\r_3 &= 11 \text{ mm} \\h_1 &= 2.6 \text{ mm} \\h_2 &= 8.7 \text{ mm} \\h_3 &= 2.6 \text{ mm}\end{aligned}$$



What about the shape?

$$r_1 = 2.05 \text{ mm}^{[1]}$$

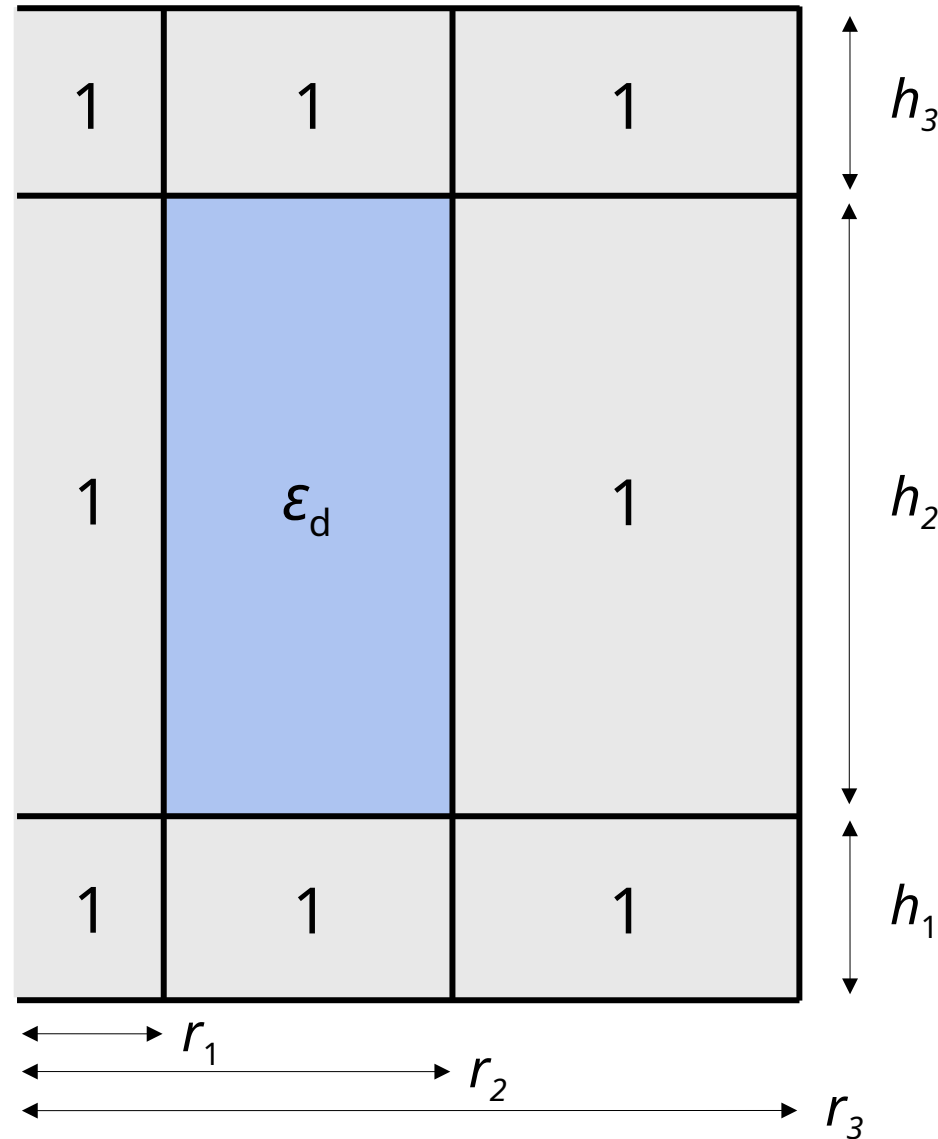
$$r_2 = 6.1 \text{ mm}$$

$$r_3 = 11 \text{ mm}$$

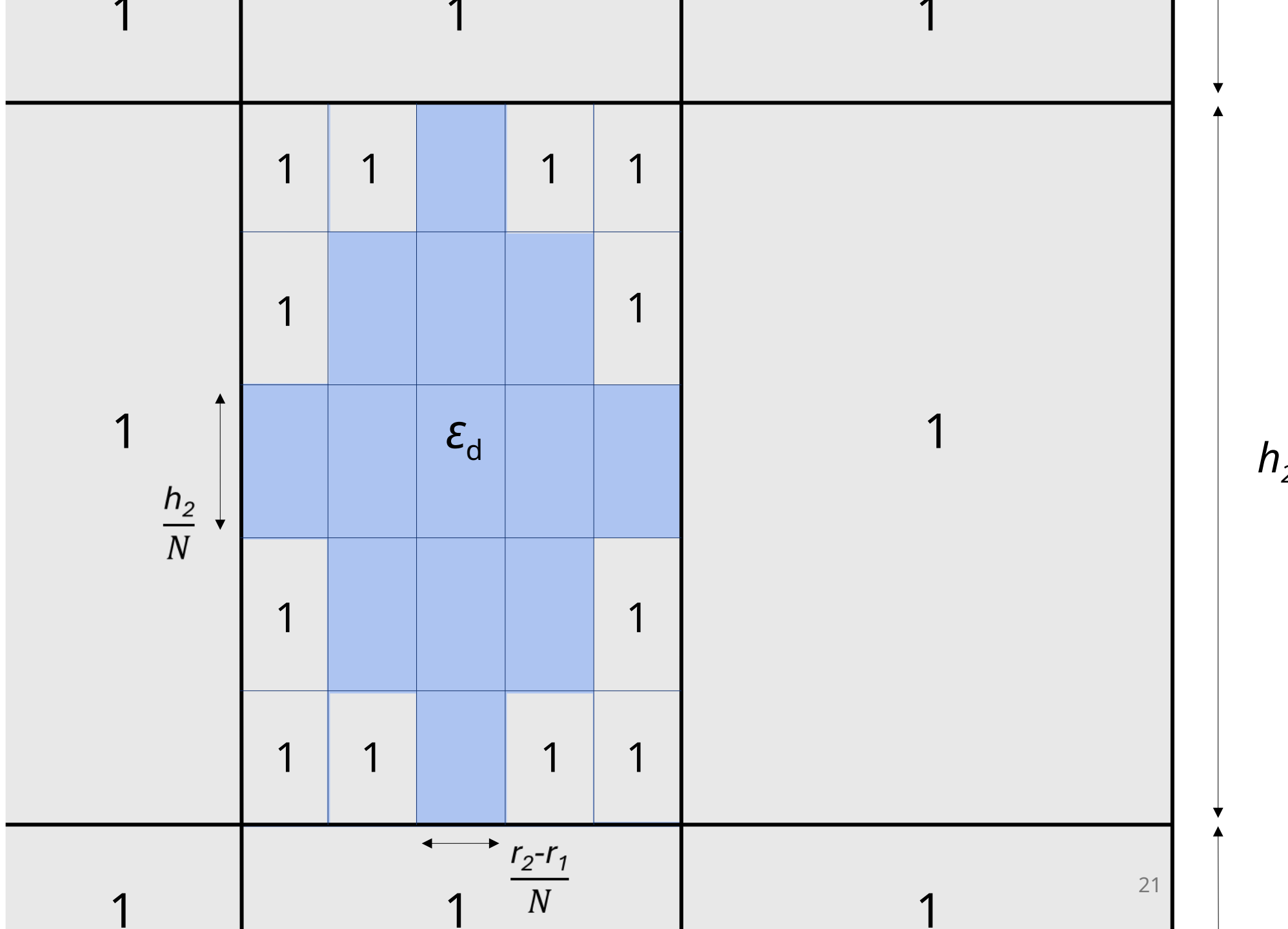
$$h_1 = 2.6 \text{ mm}$$

$$h_2 = 8.7 \text{ mm}$$

$$h_3 = 2.6 \text{ mm}$$



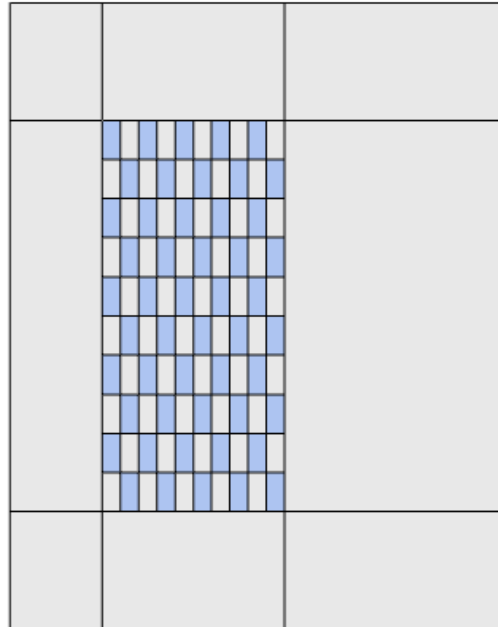
$N \times N$
(sub)
Space
Grid



Genetic Algorithms (GAs)

1. Initialisation

2. Evaluation
3. Selection
4. Crossover
5. Mutation
6. Iteration

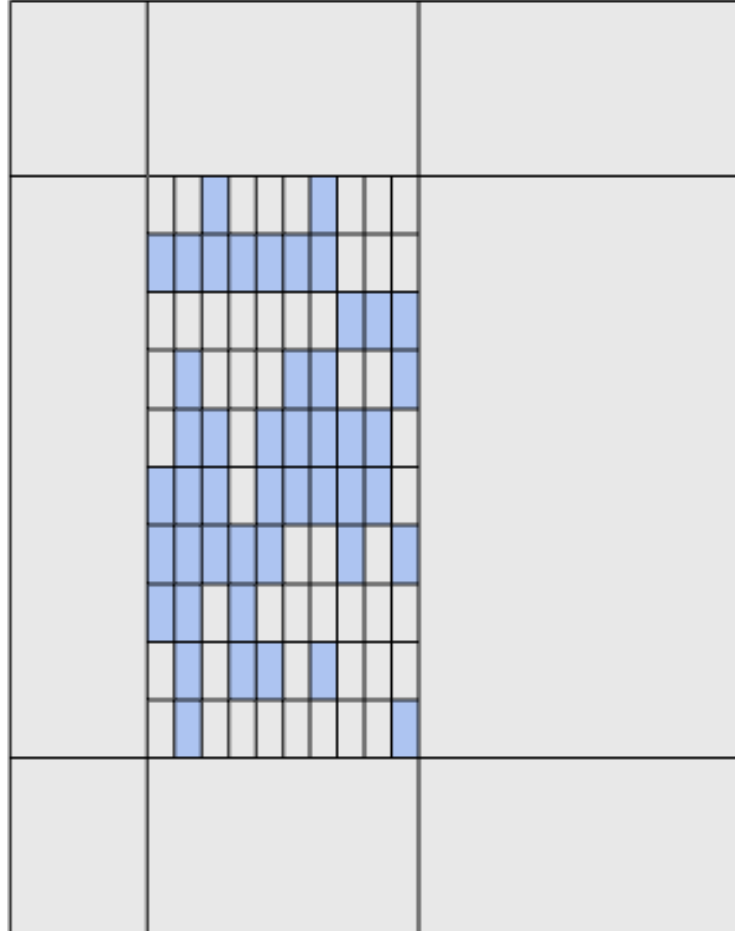


- Re-scale dimensions for 50:50 material ratio
- Randomise material distribution
- Create population of 10 random configurations:
`int populationSize = 10;`
- Set the number of generations to 100:
`int generations = 100;`

Genetic Algorithms (GAs)

1. Initialisation
- 2. Evaluation**
3. Selection
4. Crossover
5. Mutation
6. Iteration

- Run COMSOL
- Retrieve frequency



- Iteratively scale to $f_{\text{target}} = 1.45 \text{ GHz}$, using scaling factor, α :

$$\alpha = \left(\frac{f_{\text{calculated}}}{f_{\text{target}}} \right)^2$$

- Apply α to the 'global' permittivity
(Note: required damping!)
- Increasing/decreasing permittivity can be equated to decreasing/increasing the dimensions of resonator
(Note: parameters, r_x and h_x , are never changed)
- Calculate Purcell Factor

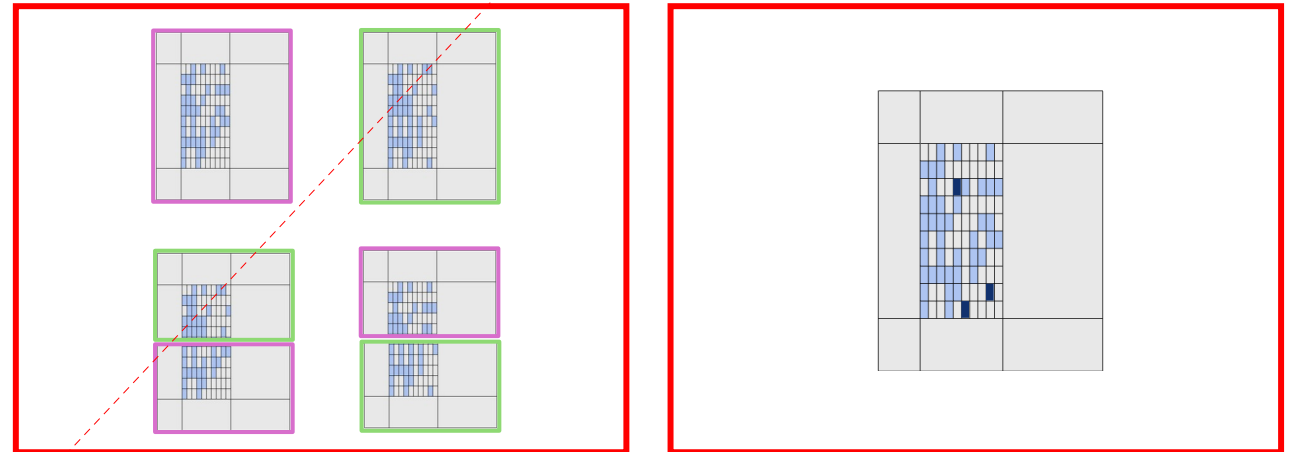
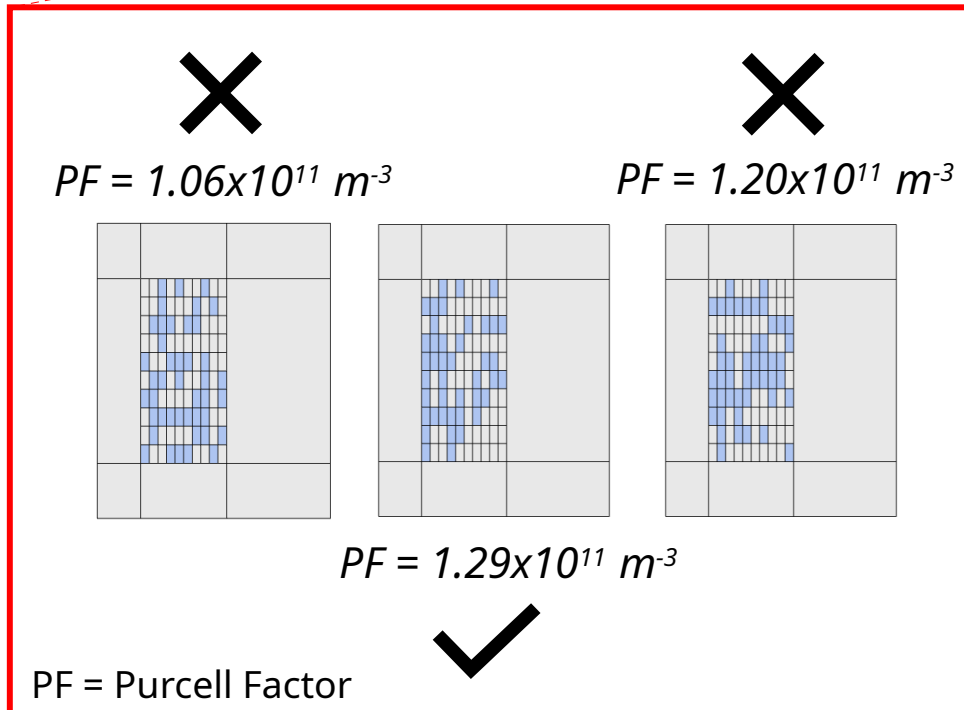
Output:

```
Iteration: 1 | Frequency: 1.4710499483913302 GHz | Alpha: 0.9625180984826076 | Purcell Factor: 1.1563318964496399E11
Iteration: 2 | Frequency: 1.451068195742074 GHz | Alpha: 0.9892089234248103 | Purcell Factor: 1.1484542545540875E11
```

Genetic Algorithms (GAs)

1. Initialisation
2. Evaluation
- 3. Selection**
4. Crossover
5. Mutation
6. Iteration

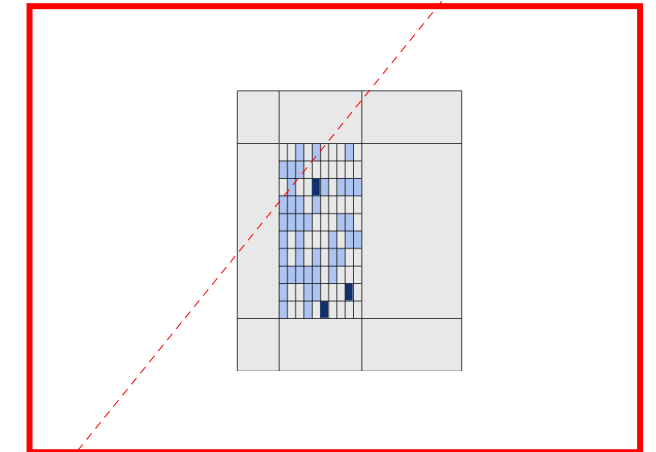
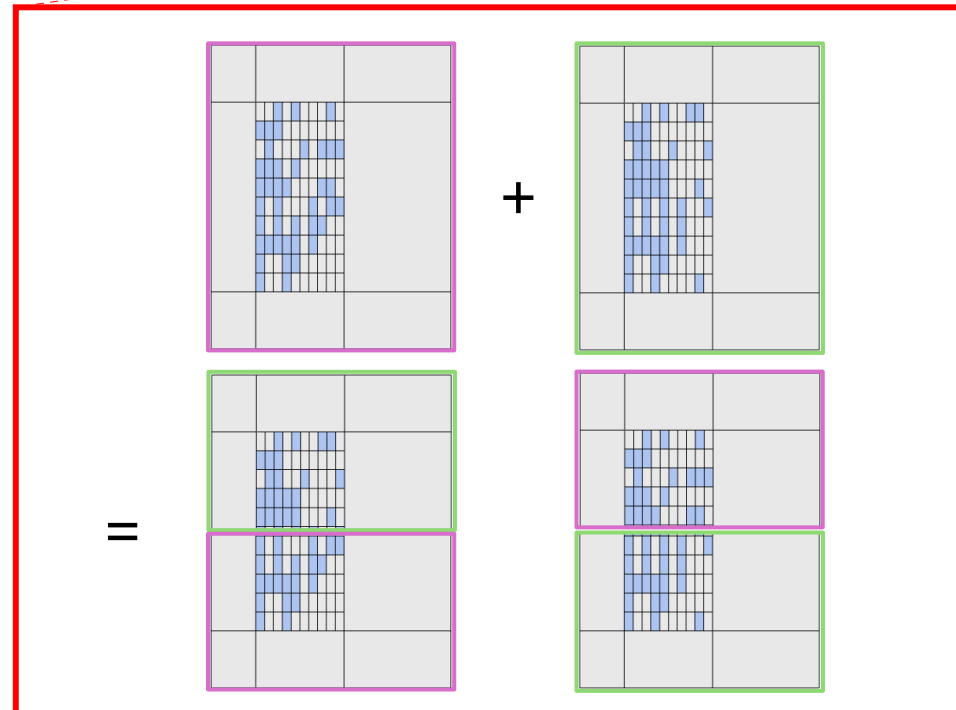
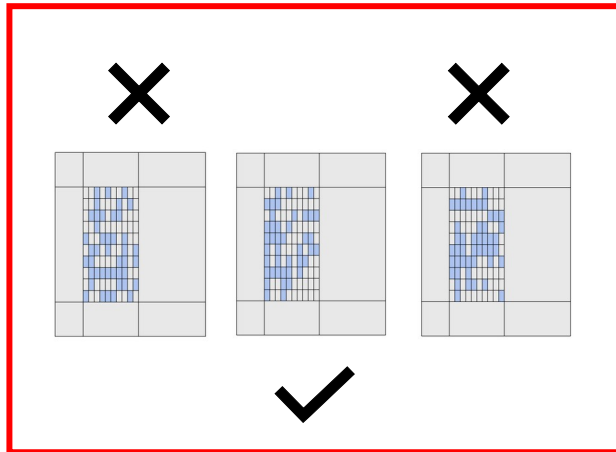
```
Engine<BitGene, Double> engine = Engine.builder(  
    fitnessFunction,  
    BitChromosome.of(100)  
)  
    .populationSize(10)  
    .selector(new TournamentSelector<>(3))  
    .alterers(new Mutator<>(0.2), new SinglePointCrossover<>(0.8))  
    .build();
```



Genetic Algorithms (GAs)

1. Initialisation
2. Evaluation
3. Selection
- 4. Crossover**
5. Mutation
6. Iteration

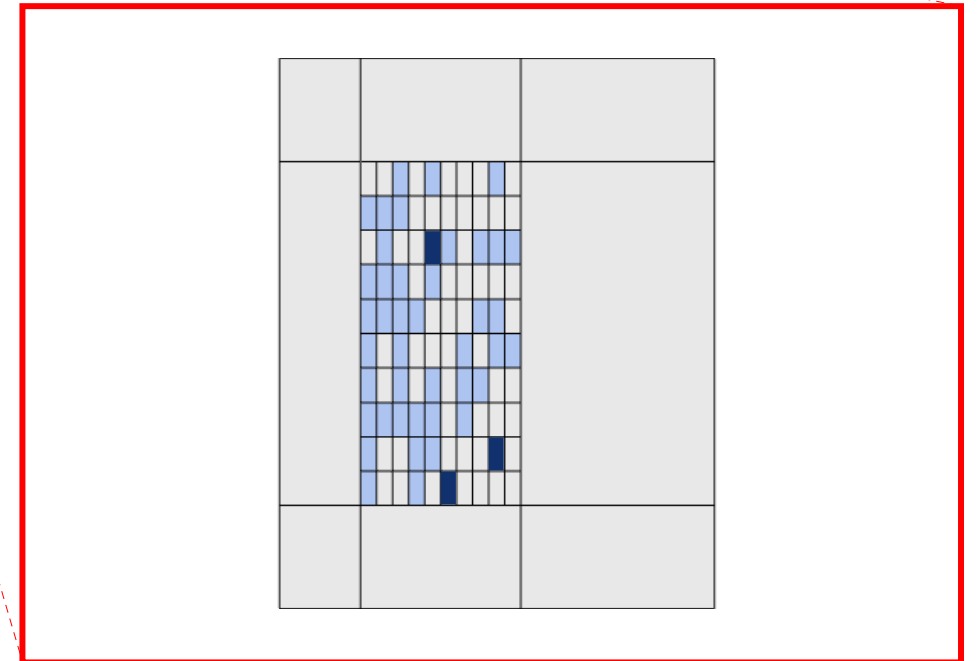
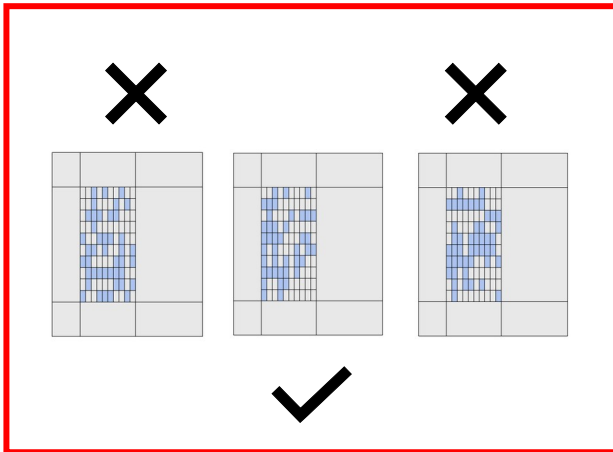
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Genetic Algorithms (GAs)

1. Initialisation
2. Evaluation
3. Selection
4. Crossover
- 5. Mutation**
6. Iteration

```
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```

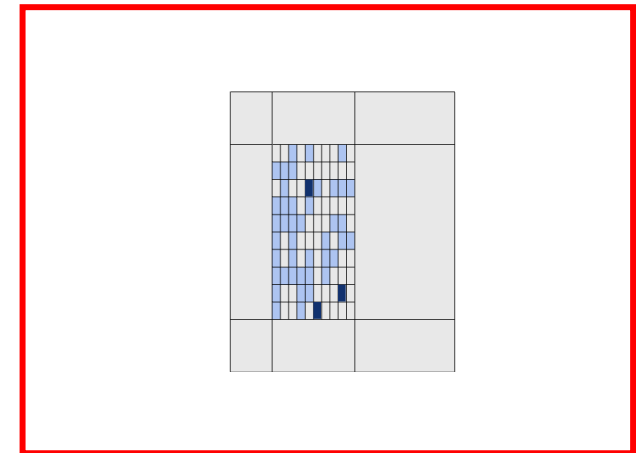
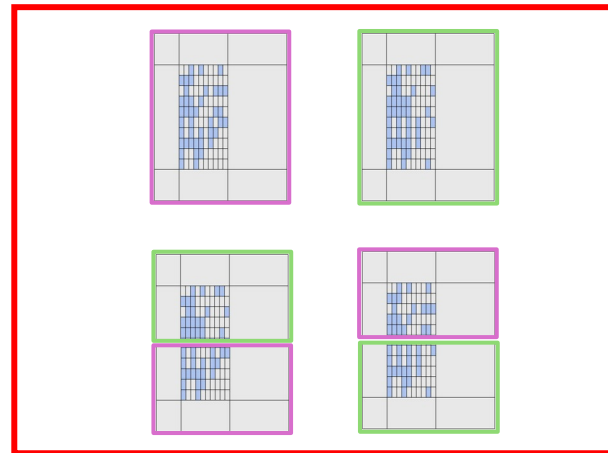
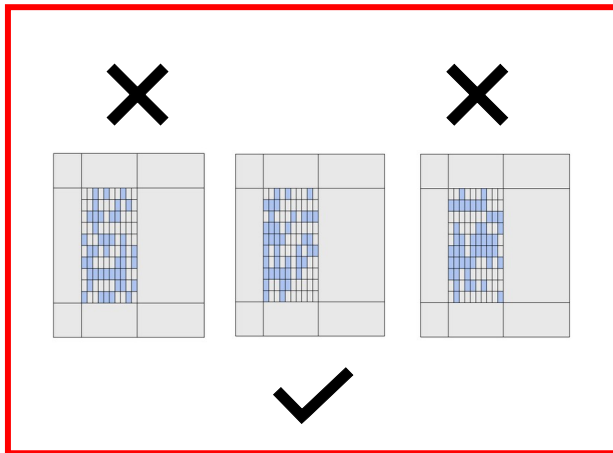


Genetic Algorithms (GAs)

1. Initialisation
2. Evaluation
3. Selection
4. Crossover
5. Mutation
- 6. Iteration**

Repeated for 100 generations

```
Engine<BitGene, Double> engine = Engine.builder(  
    fitnessFunction,  
    BitChromosome.of(100)  
)  
    .populationSize(10)  
    .selector(new TournamentSelector<>(3))  
    .alterers(new Mutator<>(0.2), new SinglePointCrossover<>(0.8))  
    .build();
```

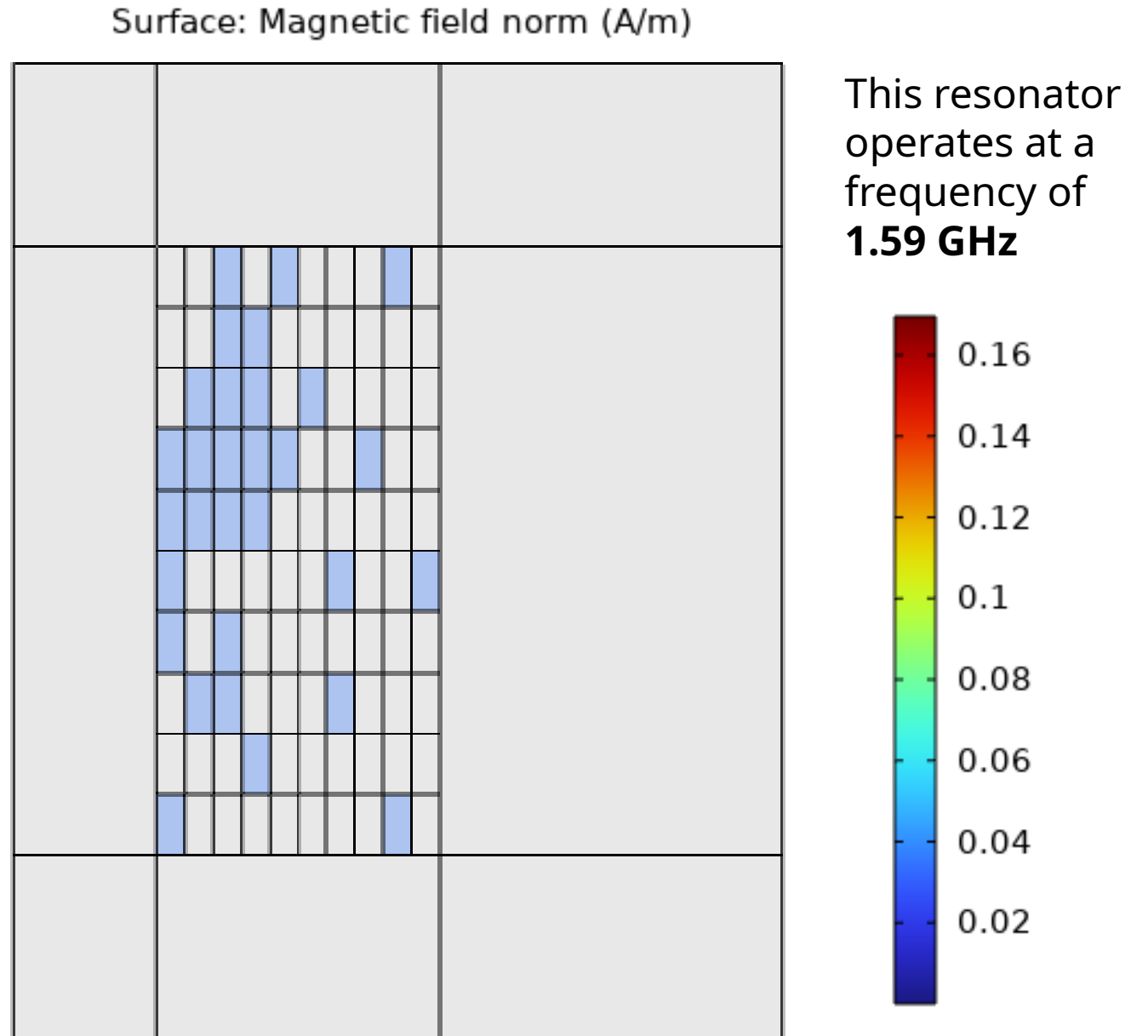
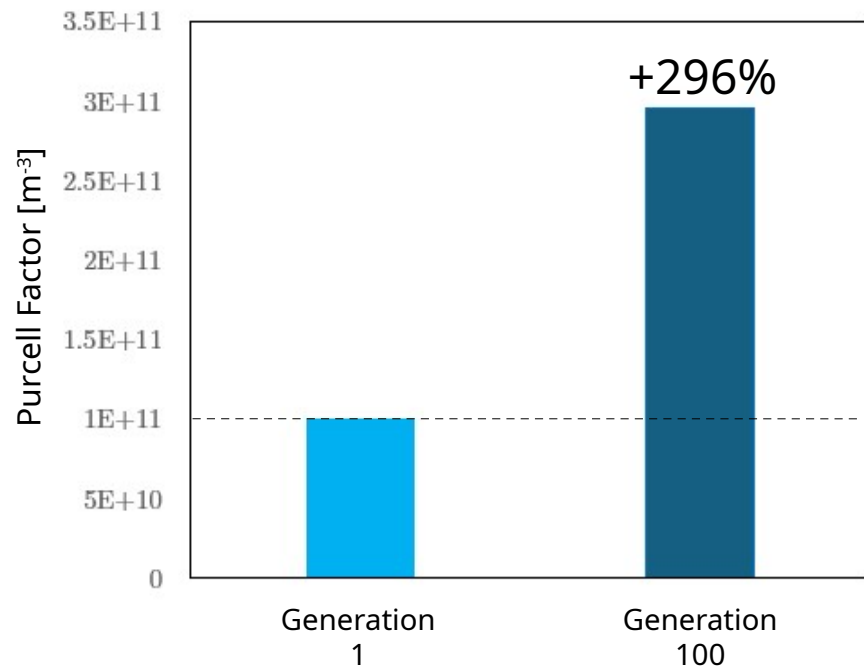


**The most “beautiful”
resonator:**

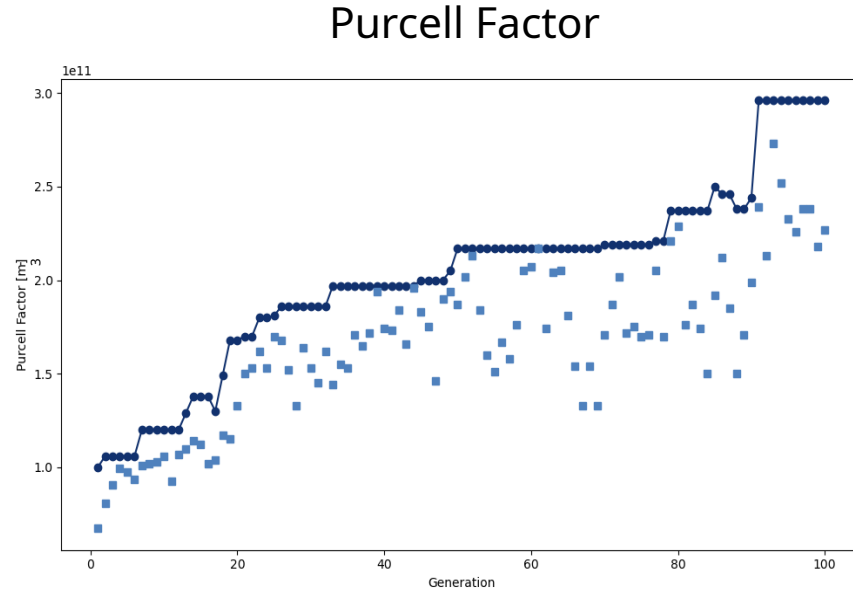
Results

Initial PF = $1.00 \times 10^{11} \text{ m}^{-3}$

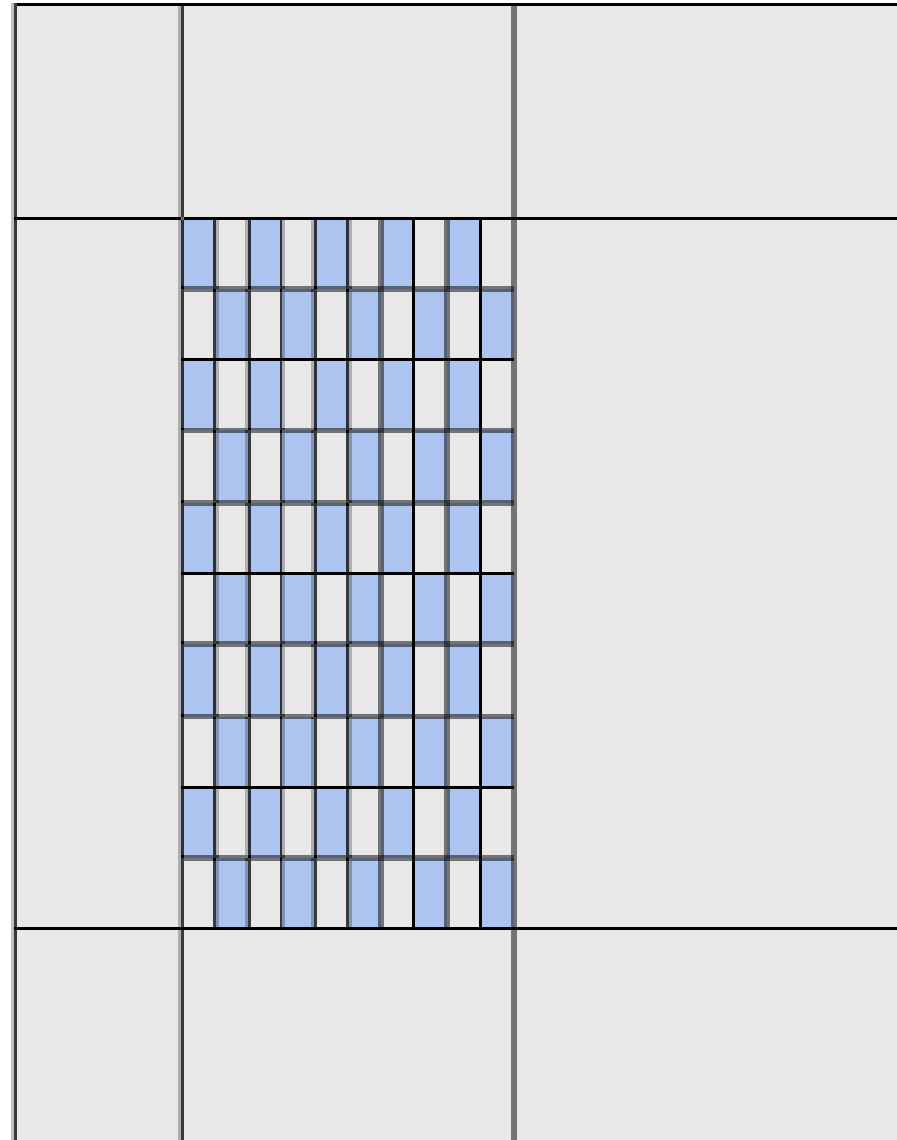
Optimised PF = **$2.96 \times 10^{11} \text{ m}^{-3}$**



Results



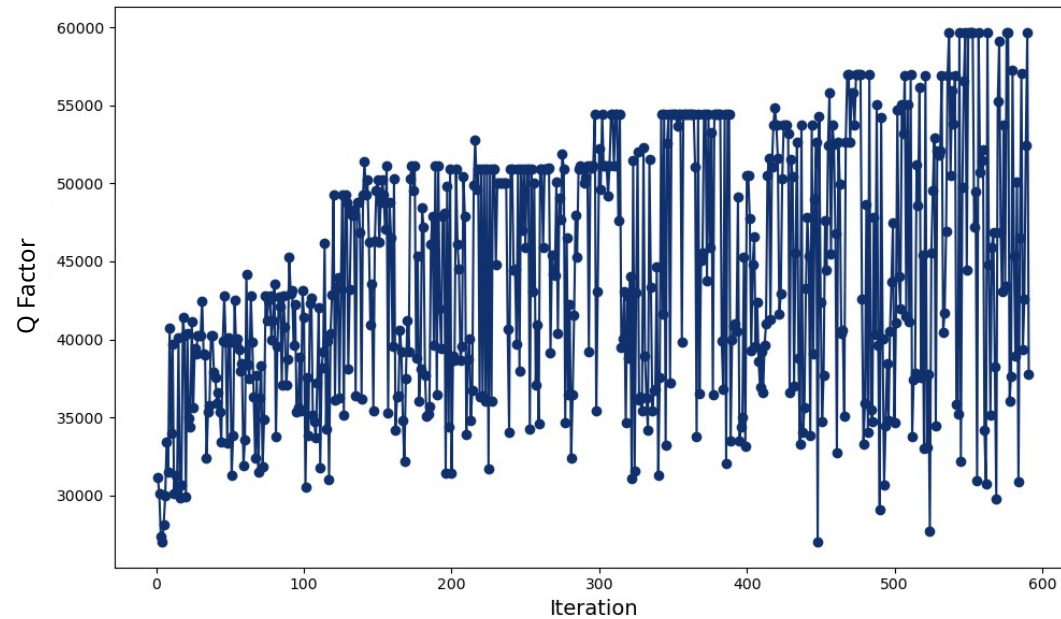
Tracking the best Purcell Factor compared to the average Purcell Factor



Animation of how the distribution between $SrTiO_3$ (blue) and air (grey) changed with generation

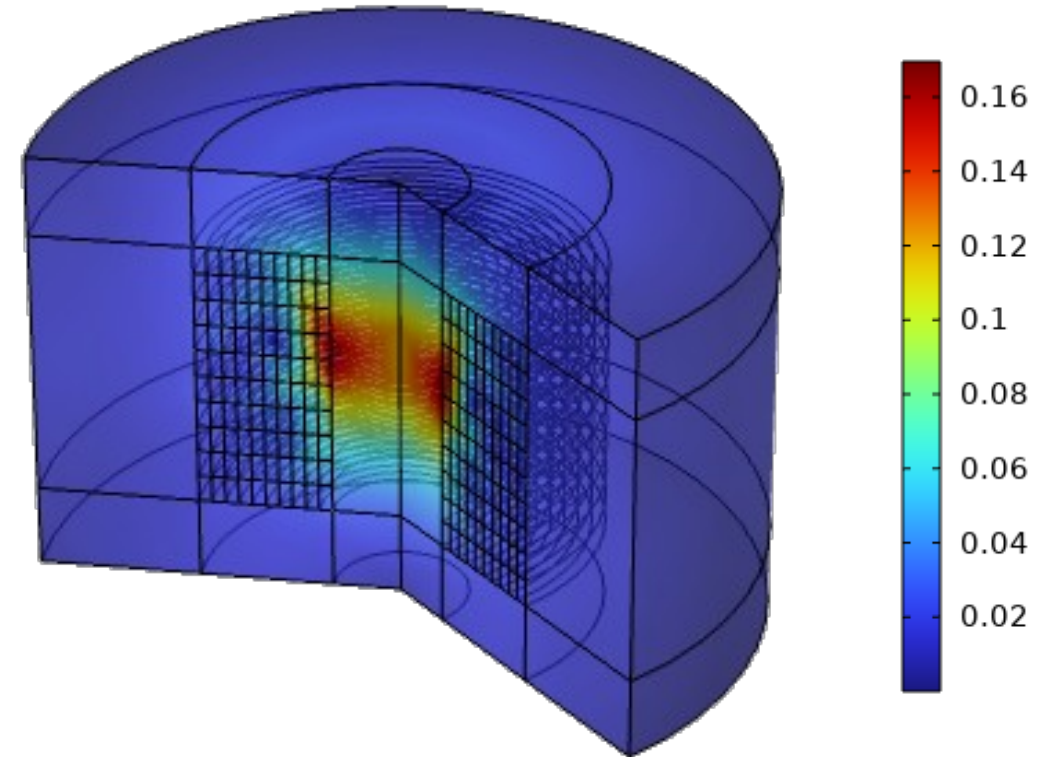
Results

Q Factor



Q Factor is driving the increase in Purcell Factor

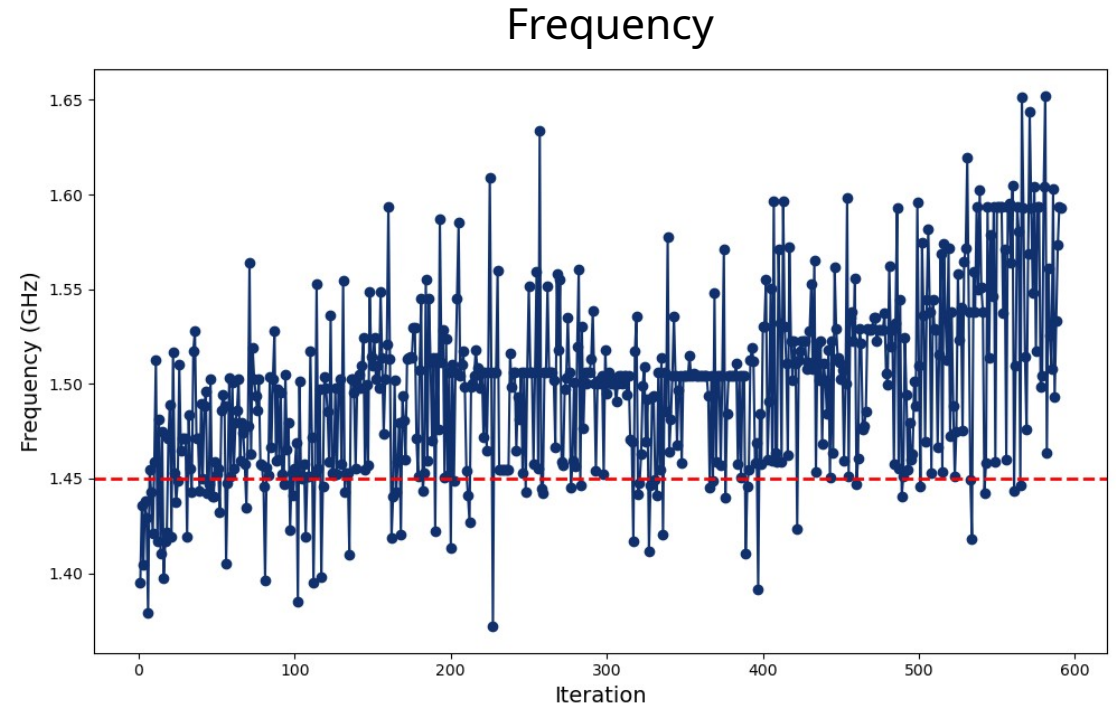
Likely due to reduction in interaction with lossy cavity walls



3D plot showing confined magnetic field

Limitations of Results

- Initial and Optimal PF not directly comparable
 - Resonance frequency different
 - Permittivities of SrTiO_3 and air different in both models
- Difficulties with manufacturing a dielectric resonator with floating annuli
- Issues with converging on 1.45 GHz
- More computational power required to see Purcell Factor plateau



Conclusion

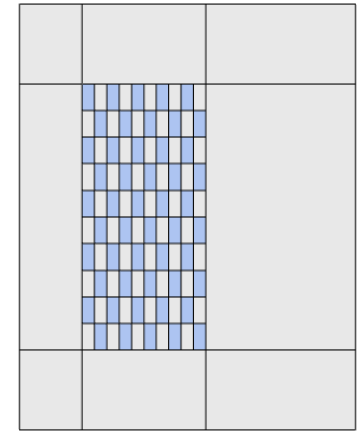
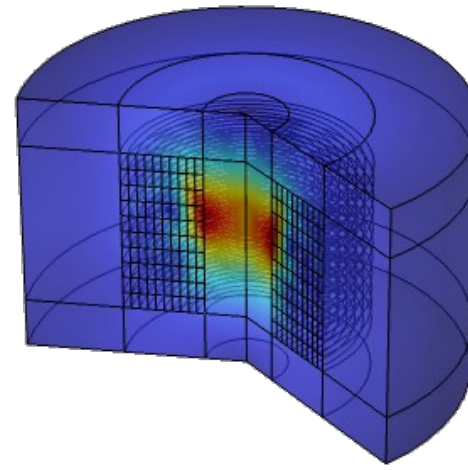
- First research that aims to optimise the shape of dielectrics resonators in MASERs.
- COMSOL Java API was implemented successfully.
- Optimal sizes for resonators were found using iterative functions
- Optimal shapes were found using GAs
- A resonator with a Purcell Factor of approximately triple the original was found using GAs.

Further Work

- Fix the resonant frequency to exactly 1.45 GHz.
- Encourage the dielectric material to 'cluster' so floating annuli are not produced.
- Use of higher performance computers
 - Ability to increase the NxN sub grid
 - More generations

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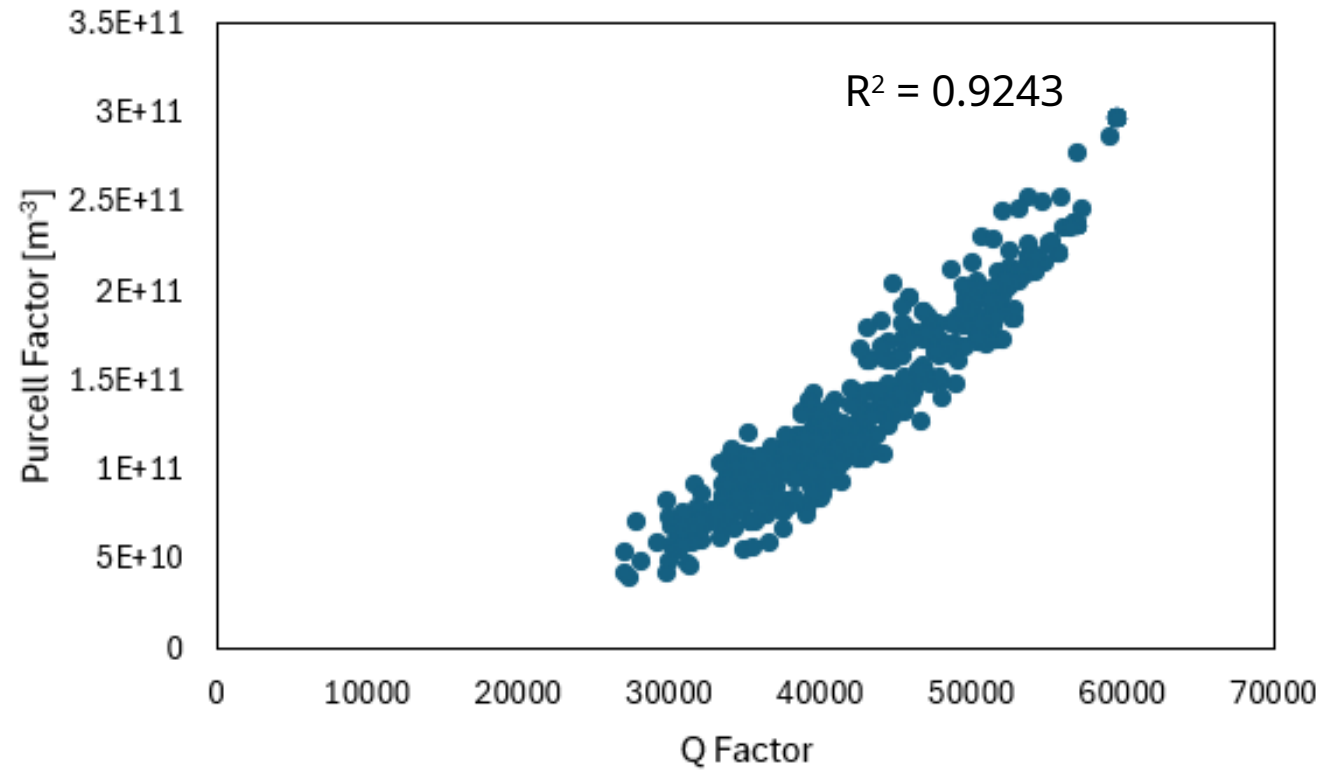


Quantum Design and Simulation of MASERs in the Strong Coupling Limit

Morgan Evans

With help from Prof. Mark Oxborrow

Appendix 1



The dielectric resonator was set up without intrinsic dielectric losses. Thus, the Q-factor – and therefore, the Purcell Factor - is primarily limited by losses from the metallic cavity boundaries.

Appendix 2

Frequency scaling based off the relationship:

$$f \propto \frac{1}{\sqrt{\epsilon_r}}$$

However, the scaling was too aggressive.
So, it was dampened:

```
double correctionFactor = Math.pow(lastFrequency / f_target, 2);  
alpha = 0.6 * alpha + 0.4 * correctionFactor;
```


Appendix 3

Quality factor:

$$Q = 2\pi \frac{\text{Energy Stored}}{\text{Energy Lost per Cycle}}$$

Mode volume:

$$V_{m,H} = \frac{\int_V \mu(\mathbf{r}) |\mathbf{H}(\mathbf{r})|^2 dV}{\max [\mu(\mathbf{r}) |\mathbf{H}(\mathbf{r})|^2]}$$

Co-operativity:

$$C = \frac{g^2}{\kappa\gamma}$$

Dielectric loss tangent:

$$\tan \delta = \frac{\epsilon''}{\epsilon'}$$

Appendix 4

Strontium titanate (SrTiO_3):

- High permittivity (~ 316)
- Low dielectric loss ($\tan \delta \approx 10^{-4}$ to 10^{-5})
- Stable and predictable microwave properties

Sapphire (Al_2O_3):

- Moderate permittivity (~ 10)
- Very low dielectric losses ($\tan \delta \approx 10^{-5}$ to 10^{-6})
- Typically used for higher GHz applications